Lecture #5: Discrete Probability Distributions

MATH 321 – Mathematical Statistics I

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Motivating Example

Example: I offer a lottery game for \$5 a play. I spin a spinner with 1–10 on it, evenly spread out. If it lands on 1–9, I keep your money. If it lands on 10, you win \$100. Let X be the payout you receive from this game.

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What is the probability mass function of the random variable X?

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What is the expected value of the random variable X, $E(X)$?

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What is the expected value of the random variable X, $E(X)$?

There are many situations like this, where we have a random experiment with two outcomes, a "success" and a "failure", with a certain probability of success. Is there a way we can take advantage of the similarities?

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Bernoulli Distribution

These kinds of random variables can be described using a Bernoulli distribution.

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A **Bernoulli random variable** is a random variable with two outcomes, a "*success*", 1; or a "*failure*", 0, with one parameter: **success probability** p.

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We write this as:

$$X \sim \text{Bernoulli}(p)$$

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We write this as:

$$X \sim \text{Bernoulli}(p)$$

What does the probability mass function of a Bernoulli(p) look like?

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PMF of the Bernoulli Distribution

The probability mass function of the Bernoulli distribution with success probability p is:

$$p_X(x) = P(X = x) = \begin{cases} p & \text{for } x = 1 \\ 1 - p & \text{for } x = 0 \end{cases}$$

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Expected Value and Variance of the Bernoulli Distribution

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Can you prove these?

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Example: Bernoulli Distribution

Example: You flip a biased coin, landing on heads 65% of the time. If the coin lands on heads, you make \$5, and if it lands on tails, you get nothing. Let X be the payout of the game.

1. What is the distribution of X?

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1. What is the distribution of X?
2. What is the expected payout?
3. What is the variance and standard deviation of the payouts?

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The Binomial Distribution

Suppose instead of flipping a single coin, we flip it 10 times, and record the number of heads, and call that variable X. How is X distributed?

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So instead of being a single Bernoulli experiment, we are performing 10 *independent* Bernoulli experiments, and recording the number of "successes".

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The Binomial Distribution

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So instead of being a single Bernoulli experiment, we are performing 10 *independent* Bernoulli experiments, and recording the number of "successes".

This is the idea behind the **binomial distribution**.

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The Binomial Distribution

The **binomial distribution** is defined as the sum of n independent and identically distributed Bernoulli random variables with success probability p.

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In formal terms:

$$X_i \overset{i.i.d.}{\sim} \text{Bernoulli}(p)$$

$$X = \sum_{i=1}^{n} X_i$$

$$X \sim \text{Binomial}(n, p)$$

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The binomial distribution has 2 parameters: the **number of trials** n and the **success probability** p.

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PMF of the Binomial Distribution

The probability mass function of a binomially distributed random variable X is:

$$p_X(x) = P(X = x) = \binom{n}{x} p^x (1-p)^{n-x},$$

for $x \in \{0, 1, 2, \ldots, n\}$.

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Cumulative Distribution Function

There are times when the probability mass function isn't the most useful way to frame a problem.

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The **cumulative distribution function (CDF)** is defined as the probability that the random variable X is less than or equal to x.

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Cumulative Distribution Function

There are times when the probability mass function isn't the most useful way to frame a problem.

The **cumulative distribution function (CDF)** is defined as the probability that the random variable X is less than or equal to x.

For a discrete random variable:

$$F_X(x) = P(X \leq x) = \sum_{k \leq x} p_X(k).$$

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Binomial Distribution Example

Example:

The probability that a patient recovers from a stomach disease is .8. Suppose 20 people are known to have contracted this disease. What is the probability that:

1. exactly 14 recover?
2. at least 10 recover?
3. at least 14 but not more than 18 recover?
4. at most 16 recover?

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Expected Value and Variance of the Binomial Distribution

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Can you prove these?

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Binomial Distribution Example

Example:

An oil exploration firm is formed with enough capital to finance ten explorations. The probability of a particular exploration being successful is .1. Assume the explorations are independent. Find the mean and variance of the number of successful explorations.

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Binomial Distribution Example

Example:

An oil exploration firm is formed with enough capital to finance ten explorations. The probability of a particular exploration being successful is .1. Assume the explorations are independent. Find the mean and variance of the number of successful explorations.

Suppose the firm has a fixed cost of \$20,000 in preparing equipment prior to doing its first exploration. If each successful exploration costs \$30,000 and each unsuccessful exploration costs \$15,000, find the expected total cost to the firm for its ten explorations.

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The Geometric Distribution

Let's consider a new game: We are going to flip the coin until we get a tails. We will pay out \$1 for each coin flip it takes until we get a tails.

- If the first coin flip is a tails (T), you get \$1.

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- If the first coin flip is a tails (T), you get \$1.
- If the first coin flip is a heads and the second coin flip is a tails (HT), you get \$2.

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- If the first coin flip is a tails (T), you get \$1.
- If the first coin flip is a heads and the second coin flip is a tails (HT), you get \$2.
- If the first two coin flips are heads and the third is a tails (HHT), you get \$3.

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- If the first coin flip is a heads and the second coin flip is a tails (HT), you get \$2.
- If the first two coin flips are heads and the third is a tails (HHT), you get \$3.
- If the first three coin flips are heads and the fourth is a tails ($HHHT$), you get \$4.

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- If the first coin flip is a heads and the second coin flip is a tails (HT), you get \$2.
- If the first two coin flips are heads and the third is a tails (HHT), you get \$3.
- If the first three coin flips are heads and the fourth is a tails ($HHHT$), you get \$4.
- And so on and so on...

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The Geometric Distribution

Let X be the payout of this game.

- What is the sample space for this game?

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The Geometric Distribution

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- What might the probability mass function look like?

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The **geometric distribution** describes the count of the number of trials until the first success, if each trial is independent and has the same probability of success p.

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The Geometric Distribution

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The **geometric distribution** describes the count of the number of trials until the first success, if each trial is independent and has the same probability of success p.

The geometric distribution has only one parameter, the **success probability** p.

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PMF of the Geometric Distribution

The probability mass function of a random variable X with a geometric distribution is:

$$p_X(x) = P(X = x) = (1-p)^{x-1}p,$$

for $x \in \{1, 2, 3, \dots\}$.

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Geometric Distribution Example

Example:

Consider the game from earlier. Suppose the coin lands on tails 70% of the time. What is the probability that:

1. the game ends on the third coin flip?
2. the game lasts 2 or fewer rounds?
3. the game lasts more than two rounds?
4. the game lasts more than four rounds, given that it has lasted more than two rounds?

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Expected Value and Variance of the Geometric Distribution

The expected value of a Geometric(p) distribution is:

$$E(X) = \frac{1}{p}.$$

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Can you prove these? (Consider a recursive definition. The textbook also uses an argument based on the derivative of the sum of a geometric series.)

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Geometric Distribution Example

Example:

Of a population of consumers, 60% are reputed to prefer a particular brand, A, of toothpaste. If a group of randomly selected consumers is interviewed, what is the mean and standard deviation of the number of consumers that must be interviewed until someone who doesn't prefer brand A is found?

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The Negative Binomial Distribution

We can further generalize the idea behind the geometric distribution. What if instead of waiting until the *first* success, we wait until the second success, or the third success, or the rth success?

- What is the sample space for this distribution?

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- What might the probability mass function look like?

The **negative binomial distribution** describes the count of the number of trials until the rth success, if each trial is independent and has the same probability of success p.

The negative binomial distribution has two parameters, the **number of successes** r and **success probability** p.

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PMF of the Negative Binomial Distribution

The probability mass function of a random variable X with a negative binomial distribution is:

$$p_X(x) = P(X = x) = \binom{x-1}{r-1}(1-p)^{x-r}p^r,$$

for $x \in \{r, r+1, r+2, \dots\}$.

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for $x \in \{r, r+1, r+2, \dots\}$.

Note that $X \sim \text{NB}(1, p)$ is the same as $X \sim \text{Geometric}(p)$.

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Expected Value and Variance of the Negative Binomial Distribution

The expected value of a $\mathrm{NB}(r, p)$ distribution is:

$$E(X) = \frac{r}{p}.$$

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Expected Value and Variance of the Negative Binomial Distribution

The expected value of a $\text{NB}(r, p)$ distribution is:

$$E(X) = \frac{r}{p}.$$

The variance of a $\text{NB}(r, p)$ distribution is:

$$V(X) = \frac{r(1-p)}{p^2}.$$

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Negative Binomial Example

A geological study indicates that an exploratory oil well should strike oil with probability .2.

1. What is the probability that the third strike comes on the seventh well drilled?
2. What assumptions did you make to obtain the answer to part (1)?
3. Find the mean and variance of the number of wells that must be drilled if the company wants to set up three producing wells.

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The Hypergeometric Distribution

Consider an urn with 8 red balls and 7 black balls in it. If I draw 5 balls from the urn, *without replacement*, what is the probability that the majority of the balls we pulled out are black?

Let Y be the number of black balls drawn from the urn in the five balls drawn.

- What is the sample space for this experiment?

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- What is the sample space for this experiment?
- What might the probability mass function look like?

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The Hypergeometric Distribution

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The **hypergeometric distribution** describes the probability of having y successes in n draws, without replacement, from a finite population of size N that contains exactly r successes.

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The **hypergeometric distribution** describes the probability of having y successes in n draws, without replacement, from a finite population of size N that contains exactly r successes.

The hypergeometric distribution has three parameters, the **total number of objects in the population,** N, the **number of "successful" objects in the population,** r, and the **number of objects drawn,** n.

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PMF of the Hypergeometric Distribution

The probability mass function of a random variable Y with a hypergeometric distribution is:

$$p_Y(y) = P(Y = y) = \frac{\binom{r}{y}\binom{N-r}{n-y}}{\binom{N}{n}},$$

where y is an integer $\{0, 1, 2, \ldots, n\}$, subject to the restrictions $y \leq r$ and $n - y \leq N - r$.

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Expected Value and Variance of the Hypergeometric Distribution

The expected value of a Hypergeometric(N, r, n) distribution is:

$$E(Y) = \frac{nr}{N}.$$

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The expected value of a Hypergeometric(N, r, n) distribution is:

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The variance of a Hypergeometric(N, r, n) distribution is:

$$V(Y) = n\left(\frac{r}{N}\right)\left(\frac{N-r}{N}\right)\left(\frac{N-n}{N-1}\right).$$

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$$V(Y) = n\left(\frac{r}{N}\right)\left(\frac{N-r}{N}\right)\left(\frac{N-n}{N-1}\right).$$

The proof requires some algebra and facts about the binomial distribution, so I won't show it here, but it's not too hard to understand.

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Hypergeometric Distribution Example

Example:

An urn contains ten marbles, of which five are green, two are blue, and three are red. Three marbles are to be drawn from the urn, one at a time without replacement.

1. What is the probability that all three marbles drawn will be green?

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Hypergeometric Distribution Example

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2. What is the probability that at most one marble drawn will be green?

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2. What is the probability that at most one marble drawn will be green?
3. What is the mean of the number of green marbles drawn?

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Hypergeometric Distribution Example

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1. What is the probability that all three marbles drawn will be green?
2. What is the probability that at most one marble drawn will be green?
3. What is the mean of the number of green marbles drawn?
4. What is the variance of the number of green marbles drawn?

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The Poisson Distribution

The manager at McDonald's called me to ask for some help modelling his sales. In particular, he wants to predict the number of customers that will come through the drive-thru in an hour, so he can properly staff the window.

Let X be the number of customers that come through the drive-thru in an hour.

- What is the sample space for this experiment?

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- What is the sample space for this experiment?
- What might the probability mass function look like?

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The Poisson Distribution

The **Poisson distribution** expresses the probability of a given number of events occurring in a fixed interval of time or space if these events occur with a known constant mean rate and independently of the time since the last event.

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The Poisson Distribution

The **Poisson distribution** expresses the probability of a given number of events occurring in a fixed interval of time or space if these events occur with a known constant mean rate and independently of the time since the last event.

The Poisson distribution has one parameter, the **average rate of events,** λ.

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PMF of the Poisson Distribution

The probability mass function of a random variable X with a Poisson distribution is:

$$p_X(x) = P(X = x) = \frac{\lambda^x \, e^{-\lambda}}{x!},$$

where x is a non-negative integer and $\lambda > 0$.

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Expected Value and Variance of the Poisson Distribution

The expected value of a Poisson(λ) distribution is:

$$E(X) = \lambda.$$

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Expected Value and Variance of the Poisson Distribution

The expected value of a Poisson(λ) distribution is:

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The variance of a Poisson(λ) distribution is:

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Expected Value and Variance of the Poisson Distribution

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$$E(X) = \lambda.$$

The variance of a Poisson(λ) distribution is:

$$V(X) = \lambda.$$

How might you prove this?

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Poisson Distribution Example

Example:

A certain type of tree has seedlings randomly dispersed in a large area, with the mean density of seedlings being approximately five per square yard.

1. What is the probability that a single one square yard region will not have any seedlings?

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Poisson Distribution Example

Example:

A certain type of tree has seedlings randomly dispersed in a large area, with the mean density of seedlings being approximately five per square yard.

1. What is the probability that a single one square yard region will not have any seedlings?
2. If a forester randomly locates ten 1-square-yard sampling regions in the area, find the probability that none of the regions will contain seedlings.

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